



SIMATS Online Programming Lab

JAVA PROGRAMMING



Bheeshma L
192511332RC



CO1-AT3-PSEUDOCODE WRITING

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QUESTIONS

Q1

Student Management System

[BL3 – Apply]

10 marks

**Q2**

2. Library Book Management

[BL3 – Apply]

10 marks

**Q3**

Employee Salary Calculation

[BL3 – Apply]

10 marks

**Q4**

Shape Area Calculator using

Polymorphism [BL3 – Apply]

10 marks

**Q5**

Vehicle Registration System

[BL3 – Apply]

10 marks

**Q6**

Bank Account Hierarchy [BL4 –

Analyze]

10 marks

**Q7**

Hospital Patient Record System

[BL4 – Analyze]

10 marks

**Q1**

Student Management System [BL3 – Apply]

10

marks

Write pseudocode to design a Student class with attributes (name, rollNo, marks) and methods to:

- Accept student details
- Calculate average marks
- Display result (Pass/Fail)

Apply encapsulation and data hiding by restricting direct access to attributes.

OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

Q8

E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

**Q9**

University Staff Hierarchy [BL4 – Analyze]
10 marks

**Q10**

Smart Home Device Controller
[BL4 – Analyze]
10 marks



10/10 answered**Class Student**

Private Name

Private rollno

Private marks

Public method acceptdetails(n,r,m[])

name=n

rollno=r

marks=m

End Method

Private Method Calculateaverage()

sum=0

For each Mark in marks:

sum=sum+marks

Return sum/Length(marks)

End Method

Public Method DisplayResults()

average=Calculateaverage()

Print "Name: ", name "| Roll.No:", rollno

If average >= 50 Then

Print "Result: Pass"

Else

Print "Result: Fail"

End IF

End Method

End Class

Save Answer

CO1-AT3-PSEUDOCODE WRITING[← Back](#)**QUESTIONS****Q1**

Student Management System

[BL3 – Apply]

10 marks

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2. Library Book Management

[BL3 – Apply]

10 marks

**Q3**

Employee Salary Calculation

[BL3 – Apply]

10 marks

**Q4**

Shape Area Calculator using

Polymorphism [BL3 – Apply]

10 marks

**Q5**

Vehicle Registration System

[BL3 – Apply]

10 marks

**Q2****2. Library Book Management [BL3 – Apply]**10
marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:

- Issue a book (change availability status)
- Return a book
- Display book details using getter methods

Demonstrate object creation for at least 3 books and simulate issue/return operations.

OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer**SIMATS**
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Q8

E-Commerce Product Catalog

[BL4 – Analyze]

10 marks

**Q9**

University Staff Hierarchy [BL4 –

Analyze]

10 marks

**Q10**

Smart Home Device Controller

[BL4 – Analyze]

10 marks



10/10 answered

```
isAvailable=True
End Method
PUBLIC METHOD getTitle() RETURN Title
PUBLIC METHOD getAuthor() RETURN Author
PUBLIC METHOD ISBN() RETURN ISBN
PUBLIC METHOD isAvailable() RETURN
isAvailable
PUBLIC METHOD issuebook()
IF getstatus() == TRUE THEN isAvailable =
FALSE
END METHOD
PUBLIC METHOD returnbook()
isAvailable=TRUE
END METHOD
PUBLIC METHOD display()
PRINT getTitle(), "|",
getAuthor(),"|",getISBN(),"| Available:",getstatus()
END METHOD
END CLASS

MAIN
CREATE b1,b2,b3 as Book
b1.init("C++","Stroustrup","111")
b2.init("JAVA","Gosling","222")
b2.init("PYTHON","Van Rossum","333")

b1.issueBook()
b2.issueBook()
b3.returnBook()

b1.display()
```

 Save Answer



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[BL3 – Apply]
10 marks

Q2

2. Library Book Management
[BL3 – Apply]
10 marks



Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks



Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks



Q5

Vehicle Registration System
[BL3 – Apply]
10 marks



Q6

Bank Account Hierarchy [BL4 –
Analyze]
10 marks



Q7

Hospital Patient Record System
[BL4 – Analyze]
10 marks



Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to:

- Calculate gross salary (basic + HRA + DA)
- Calculate net salary after deductions (PF, tax)
- Display pay slip

Apply data hiding so salary components are not directly accessible.

OOP Concepts: Encapsulation | Data Hiding |
Methods

Your Answer

Q8E-Commerce Product Catalog
[BL4 – Analyze]

10 marks

**Q9**

University Staff Hierarchy [BL4 – Analyze]

10 marks

**Q10**Smart Home Device Controller
[BL4 – Analyze]

10 marks



10/10 answered

CLASS EMPLOYEE

PRIVATE

EMPID,EMPNAME,BASICPAY,DESIGNATION

PUBLIC METHOD init(id,n,bp,d)

empid=id,name=n,basicpay=bp,designation=d

END METHOD

PUBLIC METHOD calcgross() RETURN basicpay +
(basicpay*0.70)PUBLIC METHOD calcnet() RETURN calcgross() -
(basicpay*0.72)PUBLIC METHOD displayslip()
PRINT
empid,name,designation,calcgross(),calcnet()
END METHOD

END CLASS

MAIN

CREATE emp as Employee

emp.init(101,"Alice",60000,"Dev")

emp.displayslip()

END MAIN

Save Answer

CO1-AT3-PSEUDOCODE WRITING[← Back](#)**QUESTIONS****Q1**

Student Management System

[BL3 – Apply]

10 marks

Q2**Q4****Shape Area Calculator using
Polymorphism [BL3 – Apply]**10
marks

Create a Shape base class with an overridable
calculateArea() method. Apply this to:

- Circle — area using radius

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Polymorphism [BL3 – Apply]

10 marks

Q5

Vehicle Registration System

[BL3 – Apply]

10 marks

Q6Bank Account Hierarchy [BL4 –
Analyze]

10 marks

Q7

Hospital Patient Record System

[BL4 – Analyze]

10 marks

Q8E-Commerce Product Catalog
[BL4 – Analyze]

10 marks

**Q9**

University Staff Hierarchy [BL4 – Analyze]

10 marks

**Q10**Smart Home Device Controller
[BL4 – Analyze]

10 marks



10/10 answered

CLASS SHAPE

METHOD calculatearea() RETURN 0

CLASS CIRCLE INHERITS SHAPE

r

METHOD init(rad) r = rad

OVERRIDE METHOD calculatearea() RETURN
3.14*r*r

CLASS RECTANGLE INHERITS SHAPE

l,b

METHOD init(len,brd) l = len; b=brd

OVERRIDE METHOD calculatearea() RETURN
l*b

CLASS TRIANGLE INHERITS SHAPE

b,h

METHOD init(base,ht) b=base;h=ht

OVERRIDE METHOD calculatearea() RETURN
0.5*b*h

MAIN

CREATE c as CIRCLE; c.init(5)

CREATE r as RECTANGE; r.init(4,5)

CREATE t as TRIANGLE; t.init(4,6)

PRINT c.calculatearea(), r.calculatearea(),

 Save Answer

CO1-AT3-PSEUDOCODE WRITING[← Back](#)**QUESTIONS****Q1**

Student Management System
[BL3 – Apply]
10 marks

Q2**Q5****Vehicle Registration System [BL3 – Apply]**10
marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:

- TwoWheeler — with engine capacity

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Polymorphism [BL3 – Apply]
10 marks

Q5

Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7

Hospital Patient Record System
[BL4 – Analyze]
10 marks

Q8E-Commerce Product Catalog
[BL4 – Analyze]

10 marks

**Q9**

University Staff Hierarchy [BL4 – Analyze]

10 marks

**Q10**Smart Home Device Controller
[BL4 – Analyze]

10 marks



10/10 answered

CLASS VEHICLE

regno,owner,fueltype

METHOD init(r,o,f) regno=r; owner=o; fueltype=f

METHOD displaydetails() PRINT regno, owner,
fueltype

CLASS TWOWHEELER INHERITS VEHICLE

enginecap

METHOD init(r,o,f,e) SUPER.init(r,o,f);

enginecap=e

OVERRIDE METHOD displaydetails()

SUPER.displaydetails(); PRINT enginecap

CLASS FOURWHEELER INHERITS VEHICLE

seats, ac

METHOD init(r,o,f,s,a) SUPER.init(r,o,f); seats=s;

ac=a

OVERRIDE METHOD displaydetails()

SUPER.displaydetails(); PRINT seats, ac

 Save Answer



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CO1-AT3-PSEUDOCODE WRITING

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QUESTIONS

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Shape Area Calculator using

Polymorphism [BL3 – Apply]

10 marks

**Q5**

Vehicle Registration System

[BL3 – Apply]

10 marks

**Q6**Bank Account Hierarchy [BL4 –
Analyze]

10 marks

**Q7**

Hospital Patient Record System

[BL4 – Analyze]

10 marks

**Q6**

Bank Account Hierarchy [BL4 – Analyze]

10
marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:

- Inheritance of common attributes (accNo, balance, holder)
- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.

OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

Q8E-Commerce Product Catalog
[BL4 – Analyze]

10 marks

**Q9**University Staff Hierarchy [BL4 –
Analyze]

10 marks

**Q10**Smart Home Device Controller
[BL4 – Analyze]

10 marks



10/10 answered

CLASS BANKACC

accno, bal, holder

METHOD init(a,b,h) accno=a; bal=b; holder=h

METHOD withdraw(amt) bal -= amt

CLASS SAVINGSACC INHERITS BANKACC

rate

METHOD init(a,b,h,r) SUPER.init(a,b,h); rate=r

METHOD calcInt() bal += bal*rate

OVERRIDE METHOD withdraw(amt) IF bal >= amt
THEN bal -= amt**CLASS CURRENTACC INHERITS BANKACC**

limit

METHOD init(a,b,h,l) SUPER.init(a,b,h); limit=l

OVERRIDE METHOD withdraw(amt) IF bal +
limit >= amt THEN bal -= amt Save Answer

CO1-AT3-PSEUDOCODE WRITING[← Back](#)**QUESTIONS****Q1**

Student Management System

[BL3 – Apply]

10 marks

Q2**Q7****Hospital Patient Record System [BL4 – Analyze]**10
marks

Design a Patient base class and analyze how it extends into:

- InPatient — with ward number, admission date, and

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Polymorphism [BL3 – Apply]

10 marks

Q5

Vehicle Registration System

[BL3 – Apply]

10 marks

Q6

Bank Account Hierarchy [BL4 – Analyze]

10 marks

Q7

Hospital Patient Record System

[BL4 – Analyze]

10 marks

Your Answer

Q8E-Commerce Product Catalog
[BL4 – Analyze]

10 marks

**Q9**

University Staff Hierarchy [BL4 – Analyze]

10 marks

**Q10**Smart Home Device Controller
[BL4 – Analyze]

10 marks



10/10 answered

CLASS PATIENT

PRIVATE diagnosis, prescription

name, id

METHOD init(n,i,d,p) name=n; id=i; diagnosis=d;
prescription=p

METHOD generatebill() RETURN 0

CLASS INPATIENT INHERITS PATIENT

ward, addate, dailychg, days

METHOD init(n,i,d,p,w,a,c,dy) SUPER.init(n,i,d,p);
ward=w; addate=a; dailychg=c; days=dyOVERRIDE METHOD generatebill() RETURN
dailychg * days

CLASS OUTPATIENT INHERITS PATIENT

apdate, consultfee

METHOD init(n,i,d,p,a,f) SUPER.init(n,i,d,p);
apdate=a; consultfee=fOVERRIDE METHOD generatebill() RETURN
consultfee Save Answer

CO1-AT3-PSEUDOCODE WRITING[← Back](#)**QUESTIONS****Q1**

Student Management System

[BL3 – Apply]

10 marks

Q2

2. Library Book Management

Q8**E-Commerce Product Catalog [BL4 – Analyze]**

10

marks

Analyze and design a Product class hierarchy for an e-commerce system:

- Electronics — with warranty, brand, voltage
- Clothing — with size, fabric, gender

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10 marks

Q5

Vehicle Registration System

[BL3 – Apply]

10 marks

Q6

Bank Account Hierarchy [BL4 – Analyze]

10 marks

Q7

Hospital Patient Record System

[BL4 – Analyze]

10 marks

Q8E-Commerce Product Catalog
[BL4 – Analyze]

10 marks

**Q9**

University Staff Hierarchy [BL4 – Analyze]

10 marks

**Q10**Smart Home Device Controller
[BL4 – Analyze]

10 marks



10/10 answered

CLASS Product

price

METHOD init(p) price = p

METHOD applyDiscount() RETURN price

CLASS Electronics INHERITS Product

warranty, brand, voltage

METHOD init(p, w, b, v) SUPER.init(p);

warranty=w; brand=b; voltage=v

OVERRIDE METHOD applyDiscount() RETURN
price * 0.90 // 10% discount

CLASS Clothing INHERITS Product

size, fabric, gender

METHOD init(p, s, f, g) SUPER.init(p); size=s;

fabric=f; gender=g

OVERRIDE METHOD applyDiscount() RETURN
price * 0.80 // 20% discount

CLASS Grocery INHERITS Product

expDate, weight

METHOD init(p, e, w) SUPER.init(p);

expDate=e; weight=w

OVERRIDE METHOD applyDiscount() RETURN

 Save Answer

CO1-AT3-PSEUDOCODE WRITING[← Back](#)**QUESTIONS****Q1**

Student Management System

Q9**University Staff Hierarchy [BL4 – Analyze]**10
marks**SIMATS**
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10 marks

Q4Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks**Q5**Vehicle Registration System
[BL3 – Apply]
10 marks**Q6**Bank Account Hierarchy [BL4 –
Analyze]
10 marks**Q7**Hospital Patient Record System
[BL4 – Analyze]
10 marks

without full exposure Justify the use of multilevel inheritance and explain how access modifiers enforce data hiding.

OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

Q8E-Commerce Product Catalog
[BL4 – Analyze]

10 marks

**Q9**University Staff Hierarchy [BL4 –
Analyze]

10 marks

**Q10**Smart Home Device Controller
[BL4 – Analyze]

10 marks



10/10 answered

CLASS Staff

PRIVATE id

PROTECTED basePay

METHOD calcAllowance() RETURN basePay *
0.05

CLASS NonTeachingStaff INHERITS Staff

OVERRIDE METHOD calcAllowance() RETURN
basePay * 0.10

CLASS TeachingStaff INHERITS Staff

OVERRIDE METHOD calcAllowance() RETURN
basePay * 0.15

CLASS Professor INHERITS TeachingStaff

OVERRIDE METHOD calcAllowance() RETURN
basePay * 0.30

CLASS Lecturer INHERITS TeachingStaff



Save Answer

CO1-AT3-PSEUDOCODE WRITING[← Back](#)**QUESTIONS****Q1**

Student Management System

[BL3 – Apply]

10 marks

**Q2**

2. Library Book Management

[BL3 – Apply]

10 marks

**Q3**

Employee Salary Calculation

[BL3 – Apply]

10 marks

**Q4**

Shape Area Calculator using

Polymorphism [BL3 – Apply]

10 marks

**Q10****Smart Home Device Controller [BL4 – Analyze]**10
marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation |

Inheritance

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Hospital Patient Record System

[BL4 – Analyze]

10 marks

Q8

E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

**Q9**

University Staff Hierarchy [BL4 – Analyze]
10 marks

**Q10**

Smart Home Device Controller
[BL4 – Analyze]
10 marks



10/10 answered

CLASS SmartDevice

```
METHOD executeCommand(cmd) RETURN  
"Standby"
```

CLASS SmartLight INHERITS SmartDevice

```
OVERRIDE METHOD executeCommand(cmd)  
IF cmd == "on" THEN RETURN "Light On"
```

CLASS SmartThermostat INHERITS SmartDevice

```
OVERRIDE METHOD executeCommand(cmd)  
IF cmd == "on" THEN RETURN "Heating  
Started"
```

CLASS SmartLock INHERITS SmartDevice

```
PRIVATE pin = "1234"  
OVERRIDE METHOD executeCommand(cmd)  
IF cmd == "on" THEN RETURN "Door Locked"
```

 Save Answer